Splashback and the accretion front

A companion note to *From a primordial fluctuation to an idealised rotation curve*

B. R. Parodi

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Abstract

In simulated dark-matter haloes the orbits of freshly accreted matter end at the splashback radius, where the density profile drops steeply and which separates matter that already orbits the halo from matter still falling in. Our model has a similar two-part structure: a settled galaxy inside the accretion front R_b and shells on their first infall outside it. We explain splashback for a graduate reader, place our seed in its language, and measure the splashback radius of our own seed with a one-dimensional collisionless shell code. The front and the splashback radius are related but not identical: the profile of the collisionless run drops most steeply at $1.17\,R_b$, and its orbiting matter extends to $1.6\,R_b$, in agreement with the calibration of simulated haloes. Our front is thus the inner edge of the splashback zone, close to the classical virial radius. About 15% of the orbiting matter lies outside the front, which confirms the estimate of the unrelaxed layer in the energy budget of the main paper.

1 What splashback is

Take a shell of dark matter that turns around and falls into a halo. Without collisions it does not stop: it falls through the centre, comes out on the other side, and reaches a first apocentre before falling back again. Because all shells that turn around at about the same time follow nearly the same orbit, their first apocentres pile up at one radius, where the density has a caustic. This radius is the splashback radius $r_{\rm sp}$ (Diemer & Kravtsov, 2014; Adhikari et al., 2014; More et al., 2015). Inside it, matter moves on many streams, in and out; outside it, there is only the single stream of matter on its first infall. In a smooth spherical model the caustic is a density spike; in real haloes, with a spread of orbits and non-spherical accretion, it becomes a sharp steepening of the density profile, with logarithmic slopes of -3 to -4 or steeper, located at

$$r_{\rm sp} \simeq (0.8\text{–}1.5)\, R_{\rm 200m}, \tag{1}$$

where $R_{\rm 200m}$ encloses 200 times the mean matter density. The ratio depends mainly on the mass accretion rate $\Gamma = {\rm d}\ln M/{\rm d}\ln a$: fast accretion deepens the potential while a shell is out at apocentre and pulls the apocentre in, so rapidly accreting haloes have $r_{\rm sp} \approx 0.8\text{–}1\,R_{\rm 200m}$ and slowly accreting ones $r_{\rm sp} \approx 1.5\,R_{\rm 200m}$ (More et al., 2015).

Diemer (2022) turned this into a description of the whole profile. He counts the pericentric passages of every particle and splits the density into an *orbiting* term (at least one pericentre) and an *infalling* term (none). The orbiting term is truncated sharply near $r_{\rm sp}$ and depends mostly on the accretion rate; the infalling term continues outwards and depends mostly on the environment.

2 Our seed in the language of splashback

The same two parts. In the main paper the density during growth is one function, $\rho = P(R/vt)/Gt^2$ (its Sect. 5.1): the P18 profile inside the accretion front $R_b = \xi_b vt$, $\xi_b = 1/(1{+}3\pi/2)$, and the density of the shells on their first infall outside it, with a jump by a factor 6.7 at the

front. This is the orbiting-plus-infalling decomposition of Diemer (2022), with two differences: our inner part is *settled* rather than orbiting, and our boundary is a discontinuity rather than a steep but continuous drop.

Accretion rate. Every shell of our seed has the same energy, so it turns around at $t_{\rm ta} = \pi G M / v^3 \propto M$: the halted mass grows as $M \propto t$. In an Einstein–de Sitter background, $a \propto t^{2/3}$, this is $\Gamma = \mathrm{d}\ln M / \mathrm{d}\ln a = 3/2$, a moderate accretion rate typical of dwarf haloes. In the classification of Fillmore & Goldreich (1984), whose initial overdensities fall as $\delta \propto M^{-\epsilon}$, our seed has $\epsilon = 2/3$, the case that gives an isothermal profile, $\rho \propto r^{-2}$, and hence a flat rotation curve. Bertschinger (1985) treated $\epsilon = 1$, a slower accretion ($\Gamma = 1$) with a slightly steeper profile; his collisionless solution has its outermost caustic at about a third of the current turnaround radius.

Where does our front sit? The halted shell of mass M stops at half its own turnaround radius, at $1.82\,t_{\rm ta}$, on its first infall. Measured against the shell that turns around *now*, at $r_{\rm ta} = 2vt/\pi$, the front lies at

$$\frac{R_b}{r_{\rm ta}} = \frac{\xi_b\,\pi}{2} = 0.275. \tag{2}$$

The mean density inside the front is 147 times the background (main paper, Sect. 3.3). Since $M \propto R$ outside the core, the mean density falls as R^{-2}, and the radius enclosing 200 times the background is $R_{\rm 200m} = R_b\sqrt{147/200} = 0.86\,R_b$. Our front is therefore close to the classical virial radius, just outside $R_{\rm 200m}$. With $r_{\rm sp} \simeq (1.3\text{–}1.5)\,R_{\rm 200m}$ for $\Gamma \approx 1.5$ (More et al., 2015), the splashback radius of a halo growing like our seed is expected at

$$r_{\rm sp} \simeq (1.1\text{–}1.3)\,R_b. \tag{3}$$

3 The splashback radius of our seed

To measure $r_{\rm sp}$ directly we ran the one-dimensional collisionless shell code of the main paper (its Sect. 8.1) on our seed: 1500 shells with masses from 10^{-3} to 1 (units $G = v = 1$), the common energy $-v^2/2$ and bang time, and the spin $\lambda = 0.035$ given at turnaround; shells may cross. Nine snapshots between 0.6 and 1 times the turnaround time of the outermost shell are analysed in the self-similar variable $\xi = r/vt$ (Fig. 1).

Phase space. Panel (a) shows the structure of every self-similar collisionless halo: a single cold stream of infalling matter, which turns around at $r_{\rm ta}$ and falls in with increasing speed, and inside a hot cloud of orbiting matter with radial velocities up to about $\pm 2v$. The cloud ends near $\xi \simeq 0.3$, beyond the front of our model at $\xi_b = 0.175$.

Numbers. The analysis script gives:

- the steepest drop of the density outside the front at $\xi = 0.204$, with a logarithmic slope of -4.5, i.e. at $1.17\,R_b$ or $0.32\,r_{\rm ta}$: the splashback feature as simulations define it;
- the radius enclosing 99% of the orbiting mass at $\xi = 0.276 \pm 0.009$, i.e. $1.58\,R_b$ or $0.43\,r_{\rm ta}$: the outer edge of the orbits;
- $15 \pm 2\%$ of the orbiting mass outside the front R_b.

The first number falls inside the range expected from the calibration of simulated haloes, (1.1–1.3) R_b; both lie at about a third of the current turnaround radius, like the outermost caustic of Bertschinger (1985).

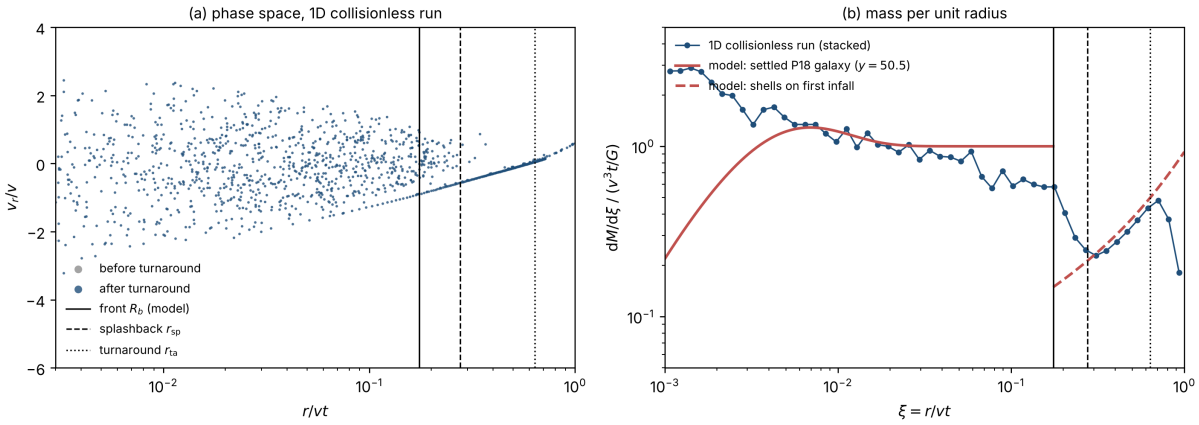


Figure 1: Our seed without halting. (a) Phase space of the last snapshot of the one-dimensional collisionless run: radial velocity against radius; grey, shells before turnaround; blue, after turnaround. The coherent stream from upper right to lower left is matter on its first infall; the cloud inside is the orbiting, multistream matter. (b) Mass per unit $\xi = r/vt$, stacked over nine snapshots (points), against the model of the main paper: the settled P18 galaxy inside the front (solid) and the cycloid infall outside it (dashed). Vertical lines: the front R_b of the model (solid), the outer edge of the orbiting matter, $r_{\rm sp}$ (dashed), and the current turnaround radius (dotted).

Profiles. Panel (b) compares the run with our model. Outside $\xi \simeq 0.3$ the two agree: the infalling stream is the same cycloid in both. Between the front and $r_{\rm sp}$ the run has orbiting matter where the model has only infall; the model's discontinuity at R_b is spread over the splashback zone. Inside, the run follows the isothermal $\mathrm{d}M/\mathrm{d}\xi \approx$ const, slightly steeper towards the centre, and has no core; this is the failure of settling in spherical symmetry discussed in the main paper.

4 What this means for our model

(A3) is a sticky version of splashback. The halting condition (A3) stops each shell on its first infall at half its turnaround radius and keeps it there. Collisionless matter does not stop; it oscillates out to the splashback radius. Both pictures conserve mass across the boundary, and both put the edge of the multistream region at a fixed fraction of the current turnaround radius, which is what makes our front move at a constant speed. Our front is the inner edge of the splashback zone, close to the virial radius; the true outer edge of the orbiting matter lies at 1.2–1.6 R_b.

The unrelaxed layer. The energy budget of the main paper (its Table 1) contains a term for the matter that has crossed the front recently and may not yet be distributed as P18 assumes; it estimated the fraction of such matter as $\xi_b \simeq 17\%$. The run gives an independent number: 15% of the orbiting matter lies outside the front. The two agree, which supports the size of that term, though not the assumed 5% error in its gravitational energy.

The infall profile. Outside the splashback zone the model's infalling density, the cycloid profile of the main paper, is not an approximation: the collisionless run follows it.

Observability. For the example galaxy of the main paper ($R_b = 84$ kpc at the end of growth) the splashback radius would lie at about 100–130 kpc, far beyond any rotation curve. Only stacked weak lensing reaches such radii; for isolated galaxies, mostly more massive than our dwarfs, it shows circular speeds that stay flat to about 300 kpc without an edge (Mistele et al., 2024).

References

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A Code

The run: `code/splashback_run.py`, which uses the shell integrator `code/shells.py` of the main paper (about four minutes); the analysis and figure: `code/splashback_fig.py`.

```
"""1D collisionless shell run of the pf2irc seed for the splashback note.
Seed: N shells, masses 1e-3..1 (units G = v = 1), common energy -1/2, common bang time;
each shell receives j = sqrt2 lambda v R_ta at turnaround (lambda = 0.035); shells may cross.
Saves 9 snapshots between 0.6 and 1 times the turnaround time of the outermost shell to
run_splash_<N>.pkl; the analysis is done by splashback_fig.py.
python3 splashback_run.py 1500 (about 4 minutes)"""
import sys, pickle, numpy as np, shells
N = int(sys.argv[1]) if len(sys.argv) > 1 else 1500
T = np.pi; snaps = list(np.linspace(0.6*T, T, 9))
out, n, tt = shells.run(N=N, Mmin=1e-3, Mmax=1.0, soft=2e-4, eta_dt=0.03, jmode='seed', lam=0.035, Tend=T,
     snaps=snaps)
pickle.dump(out, open(f"run_splash_{N}.pkl", "wb"))
print(f"N = {N}: {n} steps, {tt:.0f} s; snapshots written to run_splash_{N}.pkl")
```

```
"""Figure and numbers for the companion note on splashback.
Reads run_splash_<N>.pkl written by splashback_run.py (1D collisionless shells of the pf2irc seed,
lambda = 0.035, units G = v = 1, outermost mass 1) and compares with the halted-galaxy model of pf2irc.
python3 splashback_fig.py run_splash_1500.pkl"""
import sys, pickle, numpy as np, matplotlib.pyplot as plt
from scipy.optimize import brentq
out = pickle.load(open(sys.argv[1], "rb"))
XIB = 1/(1+1.5*np.pi)
# ---------------- model: mass per unit xi = r/vt, in units of v^3 t / G
def eta_of_xi(xi): # cycloid phase on the infalling/expanding branch for xi > xi_b
    f = lambda e: (1-np.cos(e))/(e-np.sin(e)) - xi
    return brentq(f, 1e-6, 1.5*np.pi) # outside the front: eta < 3pi/2
def mu_out(xi):
    e = eta_of_xi(xi); D = 1-np.sin(e)*(e-np.sin(e))/(1-np.cos(e))**2
    P = (e-np.sin(e))**2/(4*np.pi*(1-np.cos(e))**3*D)
    return 4*np.pi*xi**2*P
y = 50.5
xs_in = np.logspace(-3, np.log10(XIB), 200); u = 1-(1-y*xs_in/XIB)*np.exp(-y*xs_in/XIB)
mu_in = u**2 # P18: dM/dR = v^2 u^2 / G
xs_out = np.logspace(np.log10(XIB*1.0005), 0, 200); mu_o = np.array([mu_out(x) for x in xs_out])
# ---------------- simulation: stacked profile and splashback
e = np.geomspace(1e-3, 1.0, 51); acc = np.zeros(50); rsp, forb = [], []
for t, s in out.items():
    r, vr, dm, M = s["r"], s["vr"], s["dm"], s["M"]
    acc += np.histogram(r/t, e, weights=dm/t)[0]
    tta = np.pi*M
    orb = t > 2*tta # passed (approximately) the first pericentre: orbiting matter
    o = np.argsort(r[orb]); cm = np.cumsum(dm[orb][o]); cm /= cm[-1]
    rsp.append(r[orb][o][np.searchsorted(cm, 0.99)]/t) # outer edge of the orbiting matter (99% of its mass)
    forb.append(dm[orb & (r/t > XIB)].sum()/dm[orb].sum())
mu = acc/len(out)/np.diff(e); xm = np.sqrt(e[1:]*e[:-1])
rsp, forb = np.array(rsp), np.array(forb)
```

```
xta = 2/np.pi # current turnaround radius / vt
sl = np.gradient(np.log(np.maximum(mu, 1e-9)), np.log(xm)); k = (xm > XIB) & (xm < 0.5)
xsteep = xm[k][np.argmin(sl[k])]
print(f"steepest drop of dM/dxi outside the front at xi = {xsteep:.3f} (slope of rho {sl[k].min()-2:.1f}) =
    {xsteep/xta:.3f} r_ta,now")
print(f"splashback (radius enclosing 99% of the orbiting mass): r_sp = {rsp.mean():.3f} +- {rsp.std():.3f}
    vt "
     f"= {rsp.mean()/xta:.3f} r_ta,now = {rsp.mean()/XIB:.2f} R_b")
print(f"fraction of orbiting mass (t > 2 t_ta) outside the front xi_b: {forb.mean():.3f} +- {forb.std():.3f}
    ")
# ---------------- figure
tL = max(out); s = out[tL]; r, vr, M = s["r"]/tL, s["vr"], s["M"]; tta = np.pi*M
cat = np.where(tL < tta, 0, 2)
C = ["#8c8c8c", "#c0504d", "#1f4e79"]; L = ["before turnaround", "", "after turnaround"]
fig, ax = plt.subplots(1, 2, figsize=(11.5, 4.2))
for k in (0, 2):
    m_ = cat == k; ax[0].scatter(r[m_], vr[m_], s=3, c=C[k], label=L[k], alpha=0.8, lw=0)
for x_, ls_, lb in [(XIB, "-", r"front $R_b$ (model)"), (rsp.mean(), "--", r"splashback $r_{\rm sp}$"), (xta
    , ":", r"turnaround $r_{\rm ta}$")]:
    ax[0].axvline(x_, c="k", ls=ls_, lw=1, label=lb)
ax[0].set_xscale("log"); ax[0].set(xlim=(3e-3, 1.0), ylim=(-6, 4), xlabel=r"$r/vt$", ylabel=r"$v_r/v$")
ax[0].legend(frameon=False, fontsize=8, loc="lower left", markerscale=3); ax[0].set_title("(a) phase space,
    1D collisionless run", fontsize=10)
ax[1].loglog(xm[mu > 0], mu[mu > 0], "o-", ms=3, c=C[2], lw=1, label="1D collisionless run (stacked)")
ax[1].loglog(xs_in, mu_in, c=C[1], lw=1.8, label="model: settled P18 galaxy ($y=50.5$)")
ax[1].loglog(xs_out, mu_o, c=C[1], lw=1.8, ls="--", label="model: shells on first infall")
for x_, ls_ in [(XIB, "-"), (rsp.mean(), "--"), (xta, ":")]:
    ax[1].axvline(x_, c="k", ls=ls_, lw=1)
ax[1].set(xlim=(1e-3, 1.0), ylim=(0.05, 5), xlabel=r"$\xi=r/vt$", ylabel=r"$\mathrm{d}M/\mathrm{d}\xi\;/\;(v
    ^3t/G)$")
ax[1].legend(frameon=False, fontsize=8, loc="upper left"); ax[1].set_title("(b) mass per unit radius",
    fontsize=10)
fig.tight_layout(); fig.savefig("fig_splashback.png", dpi=250); print("figure: fig_splashback.png")
```

Output of the analysis:

```
steepest drop of dM/dxi outside the front at xi = 0.204 (slope of rho -4.5) = 0.321 r_ta,now
splashback (radius enclosing 99% of the orbiting mass): r_sp = 0.276 +- 0.009 vt = 0.434 r_ta,now = 1.58 R_b
fraction of orbiting mass (t > 2 t_ta) outside the front xi_b: 0.154 +- 0.016
figure: fig_splashback.png
```